

Week 03: Polymorphism and folds

Functional Programming in Lean

Learning goals

- ▶ Higher-order function arguments
- ▶ Polymorphic list functions
- ▶ `foldRight`

Core concepts

- ▶ `mapList`: elementwise function application
- ▶ `foldRight`: replacement of list constructors
- ▶ Independent element and result types

Lean example

```
def foldRight (f :  $\alpha \rightarrow \beta \rightarrow \beta$ )  
  (base :  $\beta$ ) : List  $\alpha \rightarrow \beta$   
| [] => base  
| x :: xs => f x (foldRight f base xs)
```

Lean details

- ▶ Type parameters α and β
- ▶ Curried function arguments
- ▶ Recursive result as fold accumulator

Live coding

- ▶ `mapList` by recursion
- ▶ `foldRight` by recursion
- ▶ `sum` from `foldRight`

Check

Type of `foldRight` (`·` `+` `·`) `0`

Exercise

- ▶ length from foldRight
- ▶ List identity law